# 20CYS402 DSCC [Christian Algorithm]

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Client.py:

#!/usr/bin/env python3 import socket import time from datetime import datetime import json import os

SERVER_HOST = os.environ.get('SERVER_HOST', 'time-server') SERVER_PORT = int(os.environ.get('SERVER_PORT', 8888))

## Simulated local clock drift in seconds (to vividly demonstrate synchronization)

INITIAL_DRIFT = float(os.environ.get('SIMULATED_DRIFT', '12.345678')) ROUNDS = int(os.environ.get('ROUNDS', '5')) INTERVAL = float(os.environ.get('INTERVAL', '3.0'))

class VirtualClock: def **init**(self, initial_drift: float): self.drift = initial_drift

```
def now(self) -> float:
    return time.time() + self.drift

def adjust(self, offset: float):
    self.drift += offset
```

def format_time(timestamp: float) -> str: return datetime.fromtimestamp(timestamp).strftime('%Y-%m-%d %H:%M:%S.%f')

def run_client(): print("================================================") print(" Cristian's Clock Synchronization Algorithm") print("================================================") print(f"[*] *Target Server : {SERVER_HOST}:{SERVER_PORT}") print(f"[*]* Initial Clock Drift : {INITIAL_DRIFT:+.6f} seconds") print(f"[*] Total Rounds : {ROUNDS}") print("================================================\n")

```
virtual_clock = VirtualClock(INITIAL_DRIFT)

# Wait for server container to become ready
max_retries = 15
for attempt in range(1, max_retries + 1):
    try:
        test_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        test_socket.connect((SERVER_HOST, SERVER_PORT))
        test_socket.close()
        print(f"[+] Successfully connected to time server at {SERVER_HOST}:
```

```
{SERVER_PORT}\n")
        break
    except Exception:
        if attempt == max_retries:
            print(f"[-] Could not connect to {SERVER_HOST}:{SERVER_PORT} after
{max_retries} attempts. Exiting.")
            return
        print(f"[*] Waiting for server to be reachable (attempt
{attempt}/{max_retries})...")
        time.sleep(1.5)

for i in range(1, ROUNDS + 1):
    print(f"--- [Round {i}/{ROUNDS}] ---")
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.connect((SERVER_HOST, SERVER_PORT))

        # 1. Record local time before sending request (T0)
        t0 = virtual_clock.now()
        t0_str = format_time(t0)

        # Send time request
        s.sendall(b'TIME_REQ')

        # Receive server timestamp
        response_bytes = s.recv(1024)

        # 2. Record local time upon receiving response (T1)
        t1 = virtual_clock.now()
        t1_str = format_time(t1)

        s.close()

        response = json.loads(response_bytes.decode('utf-8'))
        t_server = response['server_time']
        t_server_str = response['server_time_str']

        # 3. Calculate Round-Trip Time (RTT)
        rtt = t1 - t0

        # 4. Estimate one-way network latency
        latency = rtt / 2.0

        # 5. Calculate synchronized time according to Cristian's Algorithm
        # T_sync = T_server + (RTT / 2)
        t_sync = t_server + latency
        t_sync_str = format_time(t_sync)

        # 6. Calculate clock offset (difference between synchronized time and
client's clock at T1)
        offset = t_sync - t1

        print(f"  T0 (Client Request Sent)  : {t0_str} ({t0:.6f})")
        print(f"  T_server (Server Timestamp): {t_server_str} ({t_server:.6f})")
        print(f"  T1 (Client Response Recv) : {t1_str} ({t1:.6f})")
        print(f"  Round Trip Time (RTT)     : {rtt * 1000:.3f} ms")
        print(f"  Estimated 1-way Latency   : {latency * 1000:.3f} ms")
        print(f"  Calculated Clock Offset   : {offset:+.6f} seconds")
        print(f"  Client Time Before Sync   : {t1_str}")
```

```
        print(f"  Client Time After Sync    : {t_sync_str}")

        # Adjust virtual clock
        virtual_clock.adjust(offset)
        print(f"  Adjustment Applied        : {offset:+.6f}s")
        print(f"  Verified Client Clock Now : {format_time(virtual_clock.now())}")
        print("-------------------------------------------------\n")

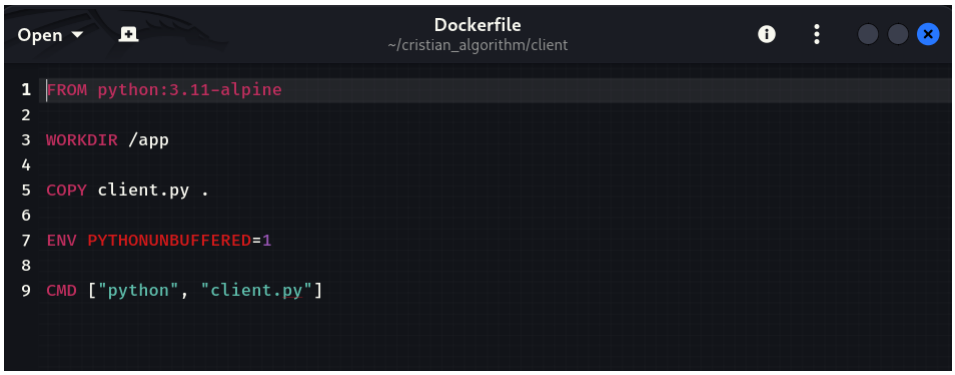
    except Exception as e:
        print(f"[-] Error during synchronization round {i}: {e}")

    if i < ROUNDS:
        time.sleep(INTERVAL)

print("[*] Cristian's Algorithm synchronization demonstration completed.")
```

if **name** == '**main**': run_client()

Client Dockerfile:




Server.py:

#!/usr/bin/env python3 import socket import time from datetime import datetime import json

HOST = '0.0.0.0' PORT = 8888

def format_time(timestamp: float) -> str: return datetime.fromtimestamp(timestamp).strftime('%Y-%m-%d %H:%M:%S.%f')

def run_server(): server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM) server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)

```
server_socket.bind((HOST, PORT))
server_socket.listen(5)
print("================================================")
print(f"[*] Cristian's Algorithm - Time Server Started")
print(f"[*] Listening on {HOST}:{PORT}")
print(f"[*] Current Server Time: {format_time(time.time())}")
print("================================================")
```

```
try:
    while True:
        conn, addr = server_socket.accept()
        with conn:
            data = conn.recv(1024)
            if not data:
                continue

            # Capture server timestamp immediately upon receiving request
            t_server = time.time()
            t_server_str = format_time(t_server)

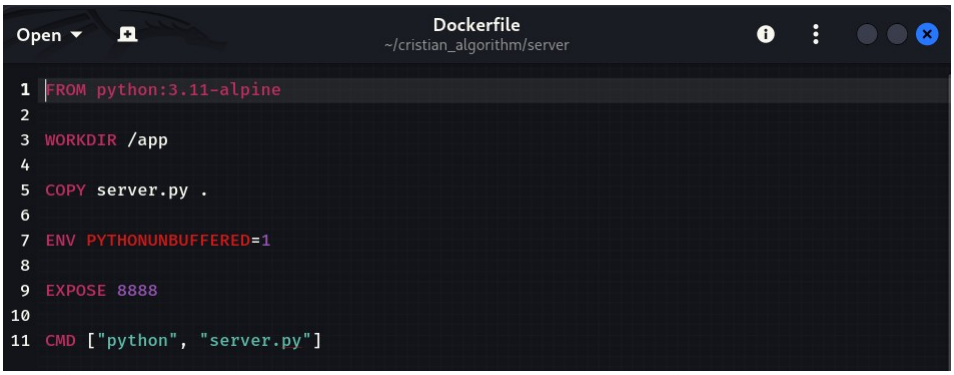
            print(f"[+] Sync request from {addr[0]}:{addr[1]}")
            print(f"    Server timestamp generated: {t_server_str}
({t_server:.6f})")

            response_payload = {
                "server_time": t_server,
                "server_time_str": t_server_str
            }

            conn.sendall(json.dumps(response_payload).encode('utf-8'))
            print(f"    Timestamp sent to client.")
except KeyboardInterrupt:
    print("\n[*] Server shutting down.")
finally:
    server_socket.close()
```

if **name** == '**main**': run_server()

Server Dockerfile:



Docker-compose.yml:

Open — docker-compose.yml — ~/cristian_algorithm

```yaml
services:
  time-server:
    build:
      context: ./server
      dockerfile: Dockerfile
    container_name: cristian-time-server
    ports:
      - "8888:8888"
    networks:
      - sync-net
    restart: unless-stopped

  time-client:
    build:
      context: ./client
      dockerfile: Dockerfile
    container_name: cristian-time-client
    depends_on:
      - time-server
    environment:
      - SERVER_HOST=time-server
      - SERVER_PORT=8888
      - SIMULATED_DRIFT=12.345678
      - ROUNDS=5
      - INTERVAL=2.0
    networks:
      - sync-net

networks:
  sync-net:
    driver: bridge
```

Compose up:

```
┌──(sanmaestro7㉿sanmaestro)-[~]
└─$ cd /home/sanmaestro7/cristian_algorithm
   docker compose up --build

[+] Building 3.6s (17/17) FINISHED
 => [internal] load local bake definitions                                0.0s
 => => reading from stdin 1.07kB                                          0.0s
 => [time-client internal] load build definition from Dockerfile          0.0s
 => => transferring dockerfile: 146B                                      0.0s
 => [time-server internal] load build definition from Dockerfile          0.0s
 => => transferring dockerfile: 159B                                      0.0s
 => [time-client internal] load metadata for docker.io/library/python:3.1 2.4s
 => [auth] library/python:pull token for registry-1.docker.io             0.0s
 => [time-server internal] load .dockerignore                             0.0s
 => => transferring context: 2B                                           0.0s
 => [time-client internal] load .dockerignore                             0.0s
 => => transferring context: 2B                                           0.0s
 => [time-client 1/3] FROM docker.io/library/python:3.11-alpine@sha256:cd 0.1s
 => => resolve docker.io/library/python:3.11-alpine@sha256:cd04730b8511de 0.1s
 => [time-client internal] load build context                             0.0s
 => => transferring context: 31B                                          0.0s
 => [time-server internal] load build context                             0.0s
 => => transferring context: 31B                                          0.0s
 => CACHED [time-client 2/3] WORKDIR /app                                 0.0s
 => CACHED [time-server 3/3] COPY server.py .                             0.0s
 => CACHED [time-client 3/3] COPY client.py .                             0.0s
 => [time-client] exporting to image                                      0.3s
 => => exporting layers                                                   0.0s
 => => exporting manifest sha256:15a28c35fcfbdceef1f8dde108e71878600e8191 0.0s
 => => exporting config sha256:7b6dfd344767ceabf0db513e1c686b2cd99b1a12ed 0.0s
 => => exporting attestation manifest sha256:2510bf8b3c795f7b5e26ae5a6f37 0.1s
 => => exporting manifest list sha256:79d5fcc947f59d0c93845f13b95ea893a63 0.0s
 => => naming to docker.io/library/cristian_algorithm-time-client:latest  0.0s
 => => unpacking to docker.io/library/cristian_algorithm-time-client:late 0.0s
 => [time-server] exporting to image                                      0.3s
 => => exporting layers                                                   0.0s
 => => exporting manifest sha256:9ff6ef615f396ca37746d8468123c0d330b5e4f0 0.0s
 => => exporting config sha256:b1ea5aef4bd338c77c598ba363c4dde46294c0f68e 0.0s
 => => exporting attestation manifest sha256:4737c444fb9b0eabd34cae10cd2c 0.1s
 => => exporting manifest list sha256:39ae18640758cba4f118094216f258c58e1 0.1s
 => => naming to docker.io/library/cristian_algorithm-time-server:latest  0.0s
 => => unpacking to docker.io/library/cristian_algorithm-time-server:late 0.0s
 => [time-client] resolving provenance for metadata file                  0.0s
 => [time-server] resolving provenance for metadata file                  0.0s
```

```
[+] up 4/4
 ✔ Image cristian_algorithm-time-server Built
 ✔ Image cristian_algorithm-time-client Built
 ✔ Container cristian-time-server       Recreated
 ✔ Container cristian-time-client       Recreated
Attaching to cristian-time-client, cristian-time-server
cristian-time-server  | ================================================
cristian-time-server  | [*] Cristian's Algorithm - Time Server Started
cristian-time-server  | [*] Listening on 0.0.0.0:8888
cristian-time-server  | [*] Current Server Time: 2026-09-22 16:07:17.393198
cristian-time-server  | ================================================
cristian-time-client  | ================================================
cristian-time-client  |   Cristian's Clock Synchronization Algorithm
cristian-time-client  | ================================================
cristian-time-client  | [*] Target Server : time-server:8888
cristian-time-client  | [*] Initial Clock Drift : +12.345678 seconds
cristian-time-client  | [*] Total Rounds  : 5
cristian-time-client  | ================================================
cristian-time-client  |
cristian-time-client  | [+] Successfully connected to time server at time-server:8888
cristian-time-client  |


cristian-time-client  | --- [Round 1/5] ---
cristian-time-server  | [+] Sync request from 172.23.0.3:46164
cristian-time-client  |   T0 (Client Request Sent)  : 2026-09-22 16:07:30.102967 (1790093250.102967)
cristian-time-client  |   T_server (Server Timestamp): 2026-09-22 16:07:17.757828 (1790093237.757828))



cristian-time-server  |     Timestamp sent to client.

cristian-time-client  |   T1 (Client Response Recv) : 2026-09-22 16:07:30.104275 (1790093250.104275)
cristian-time-client  |   Round Trip Time (RTT)     : 1.307 ms
cristian-time-client  |   Estimated 1-way Latency   : 0.654 ms
cristian-time-client  |   Calculated Clock Offset   : -12.345793 seconds
cristian-time-client  |   Client Time Before Sync   : 2026-09-22 16:07:30.104275
cristian-time-client  |   Client Time After Sync    : 2026-09-22 16:07:17.758482
cristian-time-client  |   Adjustment Applied        : -12.345793s
cristian-time-client  |   Verified Client Clock Now : 2026-09-22 16:07:17.759346
cristian-time-client  | ------------------------------------------------
```

```
cristian-time-client  |
cristian-time-client  | --- [Round 2/5] ---
cristian-time-server  | [+] Sync request from 172.23.0.3:46166
cristian-time-server  |     Server timestamp generated: 2026-09-22 16:07:19.761590 (1790093239.761590)
cristian-time-server  |     Timestamp sent to client.
cristian-time-client  |   T0 (Client Request Sent)  : 2026-09-22 16:07:19.761177 (1790093239.761177)
cristian-time-client  |   T_server (Server Timestamp): 2026-09-22 16:07:19.761590 (1790093239.761590)
cristian-time-client  |   T1 (Client Response Recv) : 2026-09-22 16:07:19.762113 (1790093239.762113)
cristian-time-client  |   Round Trip Time (RTT)     : 0.936 ms
cristian-time-client  |   Estimated 1-way Latency   : 0.468 ms
cristian-time-client  |   Calculated Clock Offset   : -0.000055 seconds
cristian-time-client  |   Client Time Before Sync   : 2026-09-22 16:07:19.762113
cristian-time-client  |   Client Time After Sync    : 2026-09-22 16:07:19.762058
cristian-time-client  |   Adjustment Applied        : -0.000055s
cristian-time-client  |   Verified Client Clock Now : 2026-09-22 16:07:19.763086
cristian-time-client  | ------------------------------------------------
cristian-time-client  |
cristian-time-client  | --- [Round 3/5] ---
cristian-time-server  | [+] Sync request from 172.23.0.3:46180
cristian-time-server  |     Server timestamp generated: 2026-09-22 16:07:21.766887 (1790093241.766887)
cristian-time-server  |     Timestamp sent to client.
cristian-time-client  |   T0 (Client Request Sent)  : 2026-09-22 16:07:21.766298 (1790093241.766298)
cristian-time-client  |   T_server (Server Timestamp): 2026-09-22 16:07:21.766887 (1790093241.766887)
cristian-time-client  |   T1 (Client Response Recv) : 2026-09-22 16:07:21.767523 (1790093241.767523)
cristian-time-client  |   Round Trip Time (RTT)     : 1.225 ms
cristian-time-client  |   Estimated 1-way Latency   : 0.612 ms
cristian-time-client  |   Calculated Clock Offset   : -0.000023 seconds
cristian-time-client  |   Client Time Before Sync   : 2026-09-22 16:07:21.767523
cristian-time-client  |   Client Time After Sync    : 2026-09-22 16:07:21.767500
cristian-time-client  |   Adjustment Applied        : -0.000023s
cristian-time-client  |   Verified Client Clock Now : 2026-09-22 16:07:21.768403
cristian-time-client  | ------------------------------------------------
cristian-time-client  |
cristian-time-client  | --- [Round 4/5] ---
cristian-time-server  | [+] Sync request from 172.23.0.3:56594
cristian-time-server  |     Server timestamp generated: 2026-09-22 16:07:23.771863 (1790093243.771863)
cristian-time-server  |     Timestamp sent to client.
cristian-time-client  |   T0 (Client Request Sent)  : 2026-09-22 16:07:23.771304 (1790093243.771304)
cristian-time-client  |   T_server (Server Timestamp): 2026-09-22 16:07:23.771863 (1790093243.771863)
cristian-time-client  |   T1 (Client Response Recv) : 2026-09-22 16:07:23.772487 (1790093243.772487)
cristian-time-client  |   Round Trip Time (RTT)     : 1.183 ms
cristian-time-client  |   Estimated 1-way Latency   : 0.592 ms
cristian-time-client  |   Calculated Clock Offset   : -0.000032 seconds
cristian-time-client  |   Client Time Before Sync   : 2026-09-22 16:07:23.772487
cristian-time-client  |   Client Time After Sync    : 2026-09-22 16:07:23.772455
cristian-time-client  |   Adjustment Applied        : -0.000032s
cristian-time-client  |   Verified Client Clock Now : 2026-09-22 16:07:23.773283
cristian-time-client  | ------------------------------------------------
```

Compose down:

